

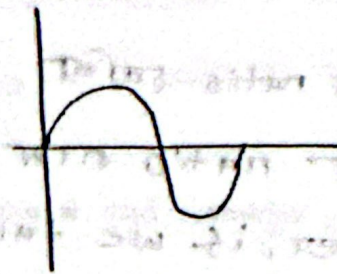
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Signal (Data): A function containing information about some phenomena.



sine function $\rightarrow$ deterministic

In a deterministic system, we can find the future value from the present value because the outcome is fixed and predictable.

Signal (Data) can be non deterministic.

The more non-deterministic (random or unpredictable) the data is, the more information it carries.

Any physical quantity (temperature, light intensity, position) varying with time (t) or space (x, y, z).

$\rightarrow f(t), f(x, y), f(x, t)$

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Signal (Data): (संकेत) Physical phenomenon that carries information

1D signal: It is a signal that varies with respect to only one independent variable, usually time or space.

2D signal: It is a signal that depends on two independent variables, usually space (x, y) .

Image $\rightarrow$ an image is a 2D signal because intensity varies in both horizontal and vertical directions.

Example: $I(x, y)$

I = image intensity at position (x, y) .

3D signal: It is a signal that depends on three independent variables, usually space (x, y, z) or space and time.

Example: Video signal $\rightarrow x(x, y, t) \rightarrow (\text{width, height, time})$

Image $f(x, y)$ means an image represented as a 2D signal or function of two spatial variables.

$x \rightarrow$ horizontal position

$y \rightarrow$ vertical position

$f(x, y) \rightarrow$ intensity (gray level or brightness) at that pixel.

f, x, y have continuous values.

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Painting $\rightarrow$ an analog signal because it contains continuous variation of intensity and color over space.

Painting $\rightarrow$ continuous $\rightarrow$ analog signal

Digital image $\rightarrow$ discrete pixels $\rightarrow$ digital signal.

Painting, medical image, traditional photography $\rightarrow$ stored physically, but not digitally.

So, we need digital image.

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Converting to Digital Image

Sampling: It means dividing the image into blocks and assigning one representative color value to each block to form the digital image.

Sampling is a lossy and non-reversible process. When we divide the image into blocks and assign one color to each block, some details are lost and the lost details cannot be recovered, we cannot get the original image back exactly.

If we increase sampling, image quality increases because more details are captured. But more pixels are stored, so more memory is required. So, higher sampling means better quality but larger file size.

Quantization: It limits the possible values of a function to a fixed set of discrete levels.

More quantization levels $\rightarrow$ smoother image $\rightarrow$ more storage needed

Fewer levels $\rightarrow$ coarser image $\rightarrow$ less storage needed.

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Storing Digital Image - Grayscale

For an image with M rows and N columns :

$x = 0, 1, 2, \dots, (N-1) \rightarrow N$ is width of the image

$y = 0, 1, 2, \dots, (M-1) \rightarrow M$ is height

Numerical array form $[s(x, y)]$

(i, j) th pixel value $[s(i, j)]$ is the image intensity at point (i, j)

↓
row
(y)

↓
column
(x)

In Python - row major order, i.e., $\text{img}[y, x]$
 $\text{img.shape}(H, W)$

classification of Digital image

Gamma rays, X-rays, ultraviolet and visible light are parts of the electromagnetic spectrum arranged from highest to lowest energy. Gamma rays have the shortest wavelength and highest frequency and energy, followed by X-rays, then ultraviolet. Visible light has longer wavelength and lower energy compared to the others, and it is the only part detectable by the human eye.

After visible light in the electromagnetic spectrum come infrared, microwave and radio waves. Infrared

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has longer wavelength and lower energy than visible light and is commonly associated with heat radiation.

Microwaves have even longer wavelengths and are used in radar and communication systems, while radio waves have the longest wavelength, lowest frequency and lowest energy, mainly used for broadcasting and wireless communication.

Reflection image

A reflection image (visible light) is formed when objects reflect visible light from a source like the sun or a lamp, and a camera captures that reflected light. This is the most common type of image used in photography and computer vision.

A reflection image (LiDAR) is formed when a LiDAR system sends laser pulses toward objects and measures the reflected signals. The reflected laser light is used to create distance and surface information, producing detailed 3D images of the environment.

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Emission image

An emission image is created when an object produces and emits its own energy. The sensor does not need an external light source because the radiation comes directly from the object itself.

i) Gamma Ray (PET imaging)

In PET imaging, a small amount of radioactive tracer is injected into the body. This tracer emits gamma rays, and the scanner detects those rays to form an image. The image shows how organs and tissues are functioning inside the body.

ii) Infrared (IR) imaging

All objects with temperature above absolute zero emit infrared radiation. A thermal camera detects this emitted heat energy and creates an image based on temperature differences. Warmer areas appear brighter or in different colors compared to cooler areas.

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Absorption image (XRAY)

An X-ray image is an absorption-based image. X-rays pass through the body and different tissues absorb X-rays differently:

Bones absorb more $\rightarrow$ appear white

Soft tissues absorb less $\rightarrow$ appear gray

Air absorbs very little $\rightarrow$ appear black

The resulting image shows contrast based on X-ray absorption, making it an absorption image.

Multimodal image - LANDSAT satellite image

A Landsat multimodal image contains multiple spectral bands (channels), each capturing different information like vegetation, water or soil. By combining 7-11 bands, the image provides detailed land, water and vegetation analysis. Each channel acts as a separate layer, giving a richer and more informative satellite image.

Image enhancement: It is the process of improving the quality of an image, often transforming a corrupted or low-quality image into a clearer, high-quality version. Example: Removing noise from a low-light photo.

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Application of Image Processing

i) Medicine

- medical image, anomaly detection, automatic disease detection

ii) Machine vision

- Quality control.
- measurement of position, orientation, sorting

iii) Military

- Battlefield awareness
- object tracing

iv) Autonomous vehicles

- self-driving cars (Tesla)
- self-exploration (NASA's Curiosity rover)